

Reference ID: RHW-25-01

Notice Type: Special Notice

Synopsis: Request for Information (RFI) for Transformational Model – Battle Management Perceive Actionable Entity (PAE) Decision Advantage Sprint for Human-Machine-Teaming (DASH)

FEDERAL AGENCY NAME: Department of Air Force, Air Force Material Command, Air Force Research Laboratory – AFRL/711th Human Performance Wing/Human Effectiveness Directorate, 2510 Fifth Street, Bldg. 840, Wright-Patterson AFB, OH 45433

1.0 GENERAL INFORMATION

1.1 THIS NOTICE IS A REQUEST FOR INFORMATION (RFI) ONLY. This RFI is issued solely for information and planning purposes and does not constitute as a Request for Proposal (RFP) or a promise to issue an RFP in the future. This request for information does not commit the Government to contract for any supply or service whatsoever. Further, the Air Force is not at this time seeking proposals and will not accept unsolicited proposals. Responders are advised that the U.S. Government will not pay for any information or administrative costs incurred in response to this RFI. All other costs associated with responding to this RFI will be solely at the interested party's expense. Not responding to this RFI does not preclude participation in any future RFI or RFP, if any is issued. If a solicitation is released, it will be synopsisized on the Beta SAM website. It is the responsibility of the potential offerors to monitor this site for additional information pertaining to this requirement. Respondents should indicate which portions of their response are proprietary and should mark accordingly. NOTE: This RFI is not associated with any previous or ongoing RFPs or Fair Opportunity Proposal Requests issued by the Department of the Air Force (DAF).

1.2 FEEDBACK. Submission is voluntary. Respondents are advised that AF/A57 and AFRL are under no obligation to provide feedback with respect to any information submitted under this RFI.

1.3 REGULATORY GUIDANCE. This publication constitutes a Request for Information (RFI) as defined in Federal Acquisition Regulation (FAR) 15.201(e), "RFIs may be used when the Government does not presently intend to award a contract, but wants to obtain price, delivery, other market information, or capabilities for planning purposes. Responses to these notices are not offers and cannot be accepted by the Government to form a binding contract."

2.0 REQUEST FOR INFORMATION (RFI)

2.1 BACKGROUND

The Deputy Secretary of Defense defines decision advantage as the ability to make better and faster decisions than an opponent. Current DAF command and control (C2) systems and processes are optimized for low intensity conflict and do not address decision advantage over the pacing challenge. Deriving effective C2 capability development toward decision advantage requires a clear understanding of what C2 decisions are. The DAF Advanced Battle Management System (ABMS) Cross Functional Team (CFT) provides a quantitative model for the Joint Warfighting Function of C2 as the Command, Plan, and Battle Management decision domains. Each of these decision domains are decomposed in the

Transformational Model for Decision Advantage (TM) using model-based systems engineering which bring these C2 decisions to light in a detailed manner.

The TM describes military decisions being made within a human-machine team (HMT), including specific choices within these domains and the battlespace information needed to inform said choices. Therefore, the TM provides foundational requirements for C2 decision-making that are universal (agnostic to time epoch, process, geography, organization, doctrine, among others). Additional information on the TMs can be found at: SAM.gov

The Transformational Model – Battle Management (TM-BM) is comprised of 13 distinct and orthogonal subfunctions (Figure 1) deduced from the following operational environment (OE) assumptions/axioms:

- No entities exist in identical states simultaneously.
- Force distributions in the OE are not uniform.
- The OE evolves over time.
- Information is incomplete and uncertain.
- Missions are difficult and resources are constrained.

In OEs where these assumptions apply (which will be the case for the pacing challenge), the TM-BM subfunctions explain the comprehensive set of battlespace information elements and decisions an HMT must make to battle manage the battlespace. Consequently, TM-BM reveals measures of performance and give structure to a design of experiments (DOE) that best characterize a battle management HMT.

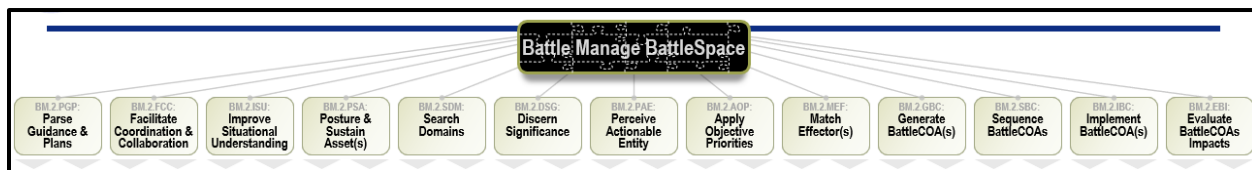


Figure 1: Transformational Model – Battle Management Subfunctions.

The resulting analysis delivers C2 *decision advantage* requirements to military capability developers and acquisition professionals through rigorous defensible analysis.

2.2 PROBLEM STATEMENT. Emerging C2 solutions focus on decision support (i.e., system interoperability, information exchange, and information presentations, i.e., battlespace awareness/display) and do not directly address decision advantage (i.e., software that machine-assists/automates battle management decision-making). To win against a peer adversary, the joint force must achieve decision advantage by equipping C2 HMTs with automation to sort through complex and high-volume battle management decisions with faster tempo and improved decision quality.

2.3 Decision Advantage Sprint for HMTs (DASH)

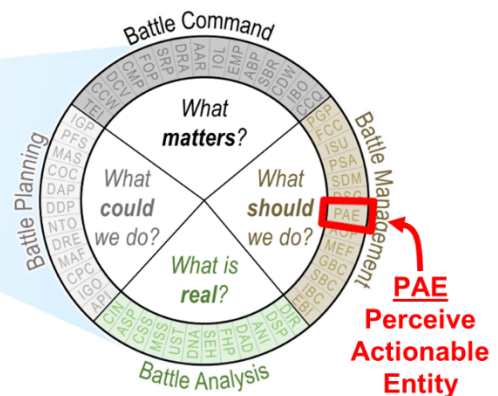
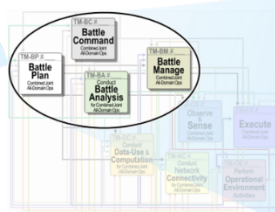
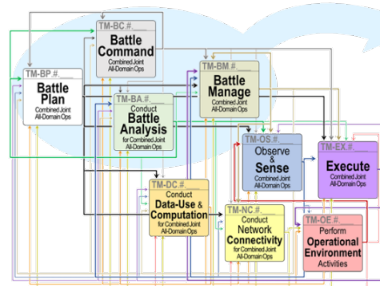
The ABMS CFT (AF/A57RB), Air Force Research Laboratory, Information Directorate (AFRL/RI), and 711 Human Performance Wing (711 HPW) are seeking information to host a Transformational Model – Battle Management “Perceive Actionable Entity” (PAE) DASH event tentatively scheduled for 31 March - 11 April 2025 at the Howard Hughes Operations (H2O) Center in Las Vegas, NV. Participants will be limited based on resource and space constraints, and selected participants will be notified by 11 March 2025.

During the DASH, participating software development teams will witness a battle management crew making PAE battle management decisions in a simulated combat environment. The participating software developers will then apply the Transformational Model (the functional description of C2 decision-making) and their creativity in developing an application/microservice that helps the humans make faster and higher quality PAE battle management decisions. Battle management warfighters will be teamed with coders to provide operational perspective and feedback. At the end of the DASH, each development team will have the opportunity to demonstrate how their software affects overall human-machine team decision making during a culminating experiment. Experiment results and the list of participants will be shared with government stakeholders. Invited participants will retain their intellectual property (IP), including any changes made over the course of the week. Government stakeholders reserve the right to analyze all collected sprint data (including software performance and technical integration insights) to derive, refine and mature functional and technical C2 software requirements.

2.4 TECHNICAL CHALLENGE

While perhaps streamlining and expediting battlespace information transport and display, current and emerging C2 systems still foist battle management decision-making on the humans. One of these decision functions is “Perceive Actionable Entity” (PAE). PAE determines possible, desirable, and permissible actions to take on something in the Operational Environment—whether it is a target (e.g., to investigate or prosecute) or a task objective (e.g., to rescue, refuel, resupply, etc.)—to achieve a specific effect, calculates the window of opportunity for those actions, and highlights conflicts between desired effects.

Transformational Model (TM) for Decision Advantage



Section A: A cover page (not to exceed a single page) identifying the company or organization, CAGE Number, street address, and the names, emails, company web page URL, and telephone numbers of the point of contact. In the case of partnerships, please provide the appropriate information for the lead POC. Include the names, emails, and phone numbers of up to five personnel who would be available, **if invited by the Government**, to attend a PAE DASH event from 31 Mar – 11 Apr 2025 at the Howard Hughes Operations (H2O) Center in Las Vegas NV. A formal sprint announcement will NOT be posted to SAM.gov, invitations will be sent directly to those organizations that are invited to attend.

State whether your company is a Small Business such as 8(a), Small Disadvantaged Business, Women-owned Business, HUBZone Small Business, Veteran-owned Business, or Service-Disabled Veteran Owned Business. Also include a statement as to whether your company is domestically or foreign owned. If foreign, indicate the country of ownership.

Also provide a short summary statement of the company's or party's experience/capabilities and a short summary of the organization's experience in the areas described above. This section is not included in the page count.

Section B: Technical Summary. The Government is assessing the current state-of-the-art and potential near-term solutions. The RFI should reflect an understanding of the PAE decision function, describe the potential product solution, and address coverage of the requirements stated in this RFI ("technical challenge") by the potential solution. As part of "understanding of the PAE decision function", include an unclassified military use case (short vignette) that demonstrates the company's/party's understanding the PAE decision function, i.e., what in the scenario constitutes a chain of events leading from knowledge about an entity of interest to knowledge about a battle effect with a time window.

The abstract shall not exceed 5 pages including figures. All abstracts may be single spaced with a font no smaller than 11 pt font. All responses to this announcement must be addressed to POCs, as discussed in Section 6.0 of this announcement. Responses should be submitted in Microsoft Office Word or PDF format.

3.2 ADDITIONAL INFORMATION. The submitted documentation and content thereof becomes the property of the U.S. Government and will not be returned. It does not commit the Government to reply to information received.

3.3 PROPRIETARY INFORMATION. This notice is part of Government market research. Information received from this request will be considered as sensitive and will be protected as such. Any company or industry proprietary information contained in responses should be clearly marked by paragraph, so that publicly releasable and proprietary information are clearly distinguished. Any proprietary information received in response to this request will be properly protected from unauthorized disclosure. The Government will not use proprietary information submitted from any one source to establish the capability and requirements for any future acquisition, so as to inadvertently restrict competition.

4.0 SPECIAL CONSIDERATIONS. Government representatives may or may not choose to meet with vendors responding to this RFI. Such discussions would only be intended to get further clarification of potential capability to meet the requirements.

The Government will review abstracts and provide email notification no later than 14 Mar 2025 of those selected for participation in a 31 Mar – 11 Apr 2025 DASH event at the H2O Center in Las Vegas NV to be held at the unclassified level. If selected and invited by the Government, no more than 4 individuals from each company may participate in the sprint week. Invitees should RSVP no later than 21 Mar 2025 with employee name, email, phone number, and clearance information to POC identified in Section 6.

5.0 SCHEDULE.

21 Feb 2025: "Ask Me Anything" on Microsoft Teams (1100 – 1300 Eastern Time)

[Join the meeting now](#) Meeting ID: 993 409 633 53 Passcode: VR9Vr9T9
+1 410-874-6750 Conference ID: 265 986 706#

7 Mar 2025: Industry abstracts due to Government by 1500 Eastern.

11 Mar 2025: Government invites selected participants.

21 Mar 2025: Industry RSVPs due.

31 Mar – 11 Apr: PAE DASH, H2O Center, Las Vegas NV.

- 31 March: Academics and initial battle management scenario run
- 1-4 April: Software development
- 7 April: Optional mid-course checkpoint with battle management scenario run
- 7-8 April: Software development
- 8-10 April: Final scenario run with software Subject Matter Experts

6.0 **POINT OF CONTACTS:**

6.1 Non-Government Advisors. Non-Government advisors may be used in the evaluation of responses and have signed non-disclosure agreements (NDAs) with the Government. Additional NDAs will not be required. Instead, a response to this RFI will be considered an implicit acknowledgement and acceptance of the fact that non-Government advisors will be assisting the Government in this process. The Government understands that information provided in response to this RFI is presented in confidence and may contain trade secret, commercial or financial information, and it agrees to protect such information from unauthorized disclosure to the maximum extent permitted or required by –

- a) 18 USC 1905 (Trade Secrets Act);
- b) 18 USC 1831 et seq. (Economic Espionage Act);
- c) 5 USC 552(b) (4) (Freedom of Information Act);
- d) Executive Order 12600 (Pre-disclosure Notification Procedures for Confidential Commercial Information); and
- e) Any other statute, regulation, or requirement applicable to Government employees

6.2 Agency Contacts: RFI responses should be submitted electronically, via email (or a DoD SAFE link can be provided) to:

Primary Technical POC:

Lt Col Timothy Wintch

ABMS CFT (AF/A57RB)

timothy.wintch@us.af.mil

Alternate POC:

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elizabeth.frost.7@us.af.mil